

Ring & Run STEM Camp 2025 – Full 3-Day Lesson Plan

Project Title: *Build a Smart AI-Powered Doorbell with Raspberry Pi*

Level: Beginner to Intermediate

Duration: 3 Days (9:00 AM – 3:00 PM Daily)

Format: Hands-on with guided modules

Outcome: Fully functioning secure IoT Smart Doorbell with integrated AI

DAY 1: Raspberry Pi Setup & Hardware Prototyping

◆ 9:00 – 9:15 | Welcome & Icebreaker

Objective: Create teams, start engagement

Activities:

- Team naming (Air Force aircraft themes)
- Warm-up discussion: “Nest vs. Ring — what’s the difference?”

Topics to Discuss:

- What is a smart doorbell?
 - Door Prize
 - What students hope to learn and build this week
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◆ 9:15 – 9:45 | STEM Project Overview

Objective: Understand full project scope

Topics to Discuss:

- Features of the smart doorbell: video, motion sensing, intercom, AI
 - Project tools: Raspberry Pi 5, sensors, web apps, OpenAI API
 - Roles of hardware, software, and networking in this project
 - IoT in the real world: smart homes, smart cities, automation
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◆ 9:45 – 10:45 | Module 1: Configure Raspberry Pi

Objective: Get Raspberry Pi up and running

Student Actions:

- Flash OS using Raspberry Pi Imager
- Set hostname, Wi-Fi, password, region
- Enable SSH and VNC
- Record IP address using terminal

Topics to Discuss:

- What is Raspberry Pi and how is it used?
- Basics of Linux and the importance in tech careers
- Why IP addresses matter in networked systems
- Introduction to SSH and remote access

Checkpoint: Confirm all teams can SSH or VNC into their Pi.

◆ 10:45 – 11:00 | Break

◆ 11:00 – 12:00 | Module 2: Build the Doorbell Hardware

Objective: Wire up the doorbell components

Student Actions:

- Use breadboard to connect PIR, LED, button
- Connect T-Cobbler and verify GPIO pins
- Connect camera module

Topics to Discuss:

- Breadboard basics (rows, columns, buses)
- Role of PIR sensor, button, LED in this project
- GPIO pinouts and pin numbering (BCM vs Board)
- Circuit protection and resistors
- Common prototyping mistakes to avoid

Checkpoint: PIR sensor triggers LED when motion is detected.

◆ 12:00 – 1:00 | Lunch + Optional Demo

◆ 1:00 – 2:00 | Module 3 Part 1: Software Environment Setup

Objective: Install required software and packages

Student Actions:

- Clone GitHub repo
- Connect Bluetooth speaker
- Adjust file permissions
- Run setup script

Topics to Discuss:

- What is Git and GitHub, and why developers use them
- Linux permissions: chmod, execute, ownership
- What the setup script does (updates, installs, audio config)

Checkpoint: Script completes successfully and speaker is configured.

◆ 2:00 – 3:00 | Module 3 Part 2: Explore Code and Run App

Objective: Understand backend and frontend structure

Student Actions:

- Navigate project files
- Open web app in browser via unsecure HTTP

Topics to Discuss:

- Backend (Python) overview: `ring_server.py`, device modes, camera logic
- Frontend (HTML/CSS/JS) overview: UI, MQTT messages, events
- MQTT basics: publish/subscribe, broker, topics
- Ports and protocols (HTTP vs HTTPS)

Checkpoint: Web app runs and camera preview displays.

DAY 2: Security & AI Integration

◆ 9:00 – 10:45 | Module 4: Secure the Connection with TLS

Objective: Add certificate-based encryption

Student Actions:

- Modify `cert_conf.sh` to match Pi IP
- Run the script to generate certs
- Import certificate into browser
- Restart Mosquitto with secure listener

Topics to Discuss:

- Difference between HTTP and HTTPS
- How TLS works (encryption, authentication, integrity)
- Public Key Infrastructure (PKI): CA, server certs, trust stores
- Why browser shows warnings with self-signed certs
- Role of OpenSSL in certificate generation

Checkpoint: Browser loads app at `https://[IP]:8001` with no warning.

◆ 10:45 – 11:00 | Break

◆ 11:00 – 12:15 | MQTT Secure Configuration

Objective: Ensure messages are encrypted over MQTT

Student Actions:

- Run `MQTT_listener_secure_conf.sh`
- Restart services
- Confirm MQTT works over secure port

Topics to Discuss:

- Why MQTT needs security (data interception, spoofing)
- Secure ports (1883 vs 8883)
- Restarting services and checking logs for errors

Checkpoint: MQTT messages pass securely with no errors.

◆ 12:15 – 1:00 | Lunch + Fluffy

◆ 1:00 – 2:45 | Module 5: Make It Smart with AI (OpenAI)

Objective: Add GPT-4o image processing

Student Actions:

- Decode hidden API key (steganography exercise)
- Add key to `.env` file
- Modify `handleGPTRequest()`
- Test "Ask GPT" button

Topics to Discuss:

- What is Artificial Intelligence and what can it do?
- OpenAI's GPT-4o: vision + language model
- API keys: what they are and how they authorize use
- Difference between API key and TLS cert
- How image data is encoded and sent to OpenAI

Checkpoint: GPT returns a description of the doorbell's camera view.

DAY 3: Customization & Presentations

◆ 9:00 – 11:00 | Module 6: Doorbell Customization

Objective: Personalize and expand the project

Suggested Activities:

- Modify CSS for new themes (team colors)
- Add custom doorbell ringtones (.mp3)
- Brainstorm and prototype cloud video upload (AWS, Azure, etc.)
- Experiment with different AI prompts

Topics to Discuss:

- What makes an IoT project scalable or market-ready?
- Pros/cons of cloud vs local storage
- OpenAI customization (system prompts, prompt engineering)
- Alternative AI models and platforms

Checkpoint: Each team implements one meaningful change.

◆ **11:00 – 11:15 | Break**

◆ **11:15 – 12:00 | Final Touches & Practice Presentations**

Objective: Prepare for sharing final work

Topics to Discuss:

- What problem does your doorbell solve?
 - What technical skills did you gain?
 - If you could do this again, what would you improve?
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◆ **12:00 – 1:45 | Team Presentations**

Objective: Share what was learned

Each Team Presents (5–7 minutes):

- What did you build and customize?
 - What challenges did you overcome?
 - What inspired you most?
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◆ **1:45 – 2:45 | Raspberry Pi Door Prize Drawing + Feedback**

Topics to Discuss:

- What other IoT projects could you try next?
 - What careers use these skills?
 - Where can you keep learning?
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◆ **2:45 – 3:00 | Camp Wrap Up**

- Final Q&A
- Complete survey

- Thank students and invite them to share their ideas online